

Power Platform for the Perspective from a Business Central Developer





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Some challenges in the world of software development



Repeatability

Build Reusable Assets

One-Time Effort, Many Sales

Scalable Delivery

- Fast, efficient

Maintainability

- Automated Upgrades
- Hotfixes

Standardized foundation

- BC & PP Platform

Maximize ROI

Maintainability

Automation

- CI/CD
- Validation
- Upgrades

Environment Accessibility (GDAP)

Product feedback

- Telemetry



Development Methodology

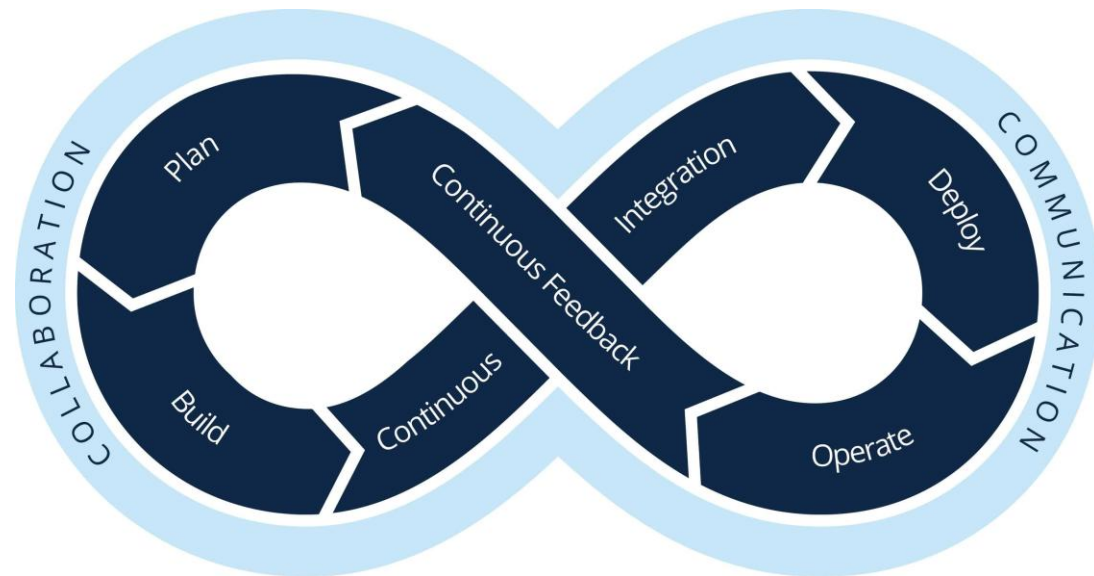
Dev environments

- DEV / QA / UAT / PROD / TRAINING

Versioning & branching

VNext

- For the product
- For your apps



Customization

Customers are never 100% happy with an out-of-the-box product.

- They have "unique" processes
- Improve efficiency
- Meet their industry needs
- Enhance user experience
- ...

VAR vs ISV

VAR

ISV

What?

Value Added Reseller
Adds value through services to existing products

Independent Software Vendor
Develops and sells its own software products

Focus

Service Oriented:
Consultancy, training, support.
Customer Customization

Product-oriented: building reusable, scalable software solutions
Maintenance.

Key Trait

Enhances and delivers someone else's product.

Owns and markets its own IP

Business Central in a nutshell

What is Business Central?

*Microsoft Dynamics 365 Business Central is a **cloud-based, all-in-one ERP** (Enterprise Resource Planning) solution designed for **small to medium-sized businesses***

*Its **purpose** is to streamlines operations—finance, sales, inventory, purchasing, and more—in a single platform.*

*It is built on **modular** architecture and is highly **customizable** and **extensible**.*

Technical Backbone

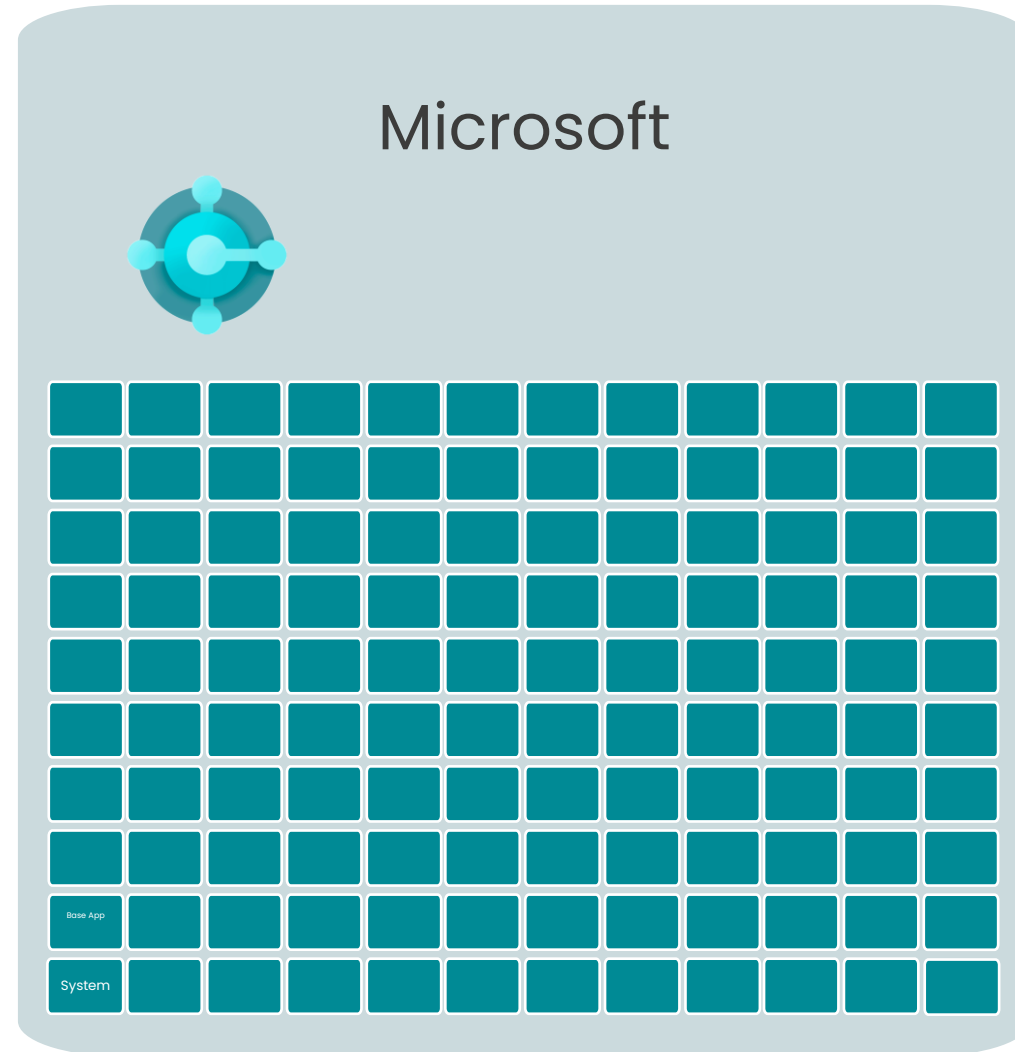
Built on AL

Extensibility:

- The ability to create apps in AL
- Microsoft
 - creates base apps (in AL)
 - provides many "events" to be able to tap into from other apps
- Partners
 - Create apps as products (AppSource) or specific for customers (PTE)

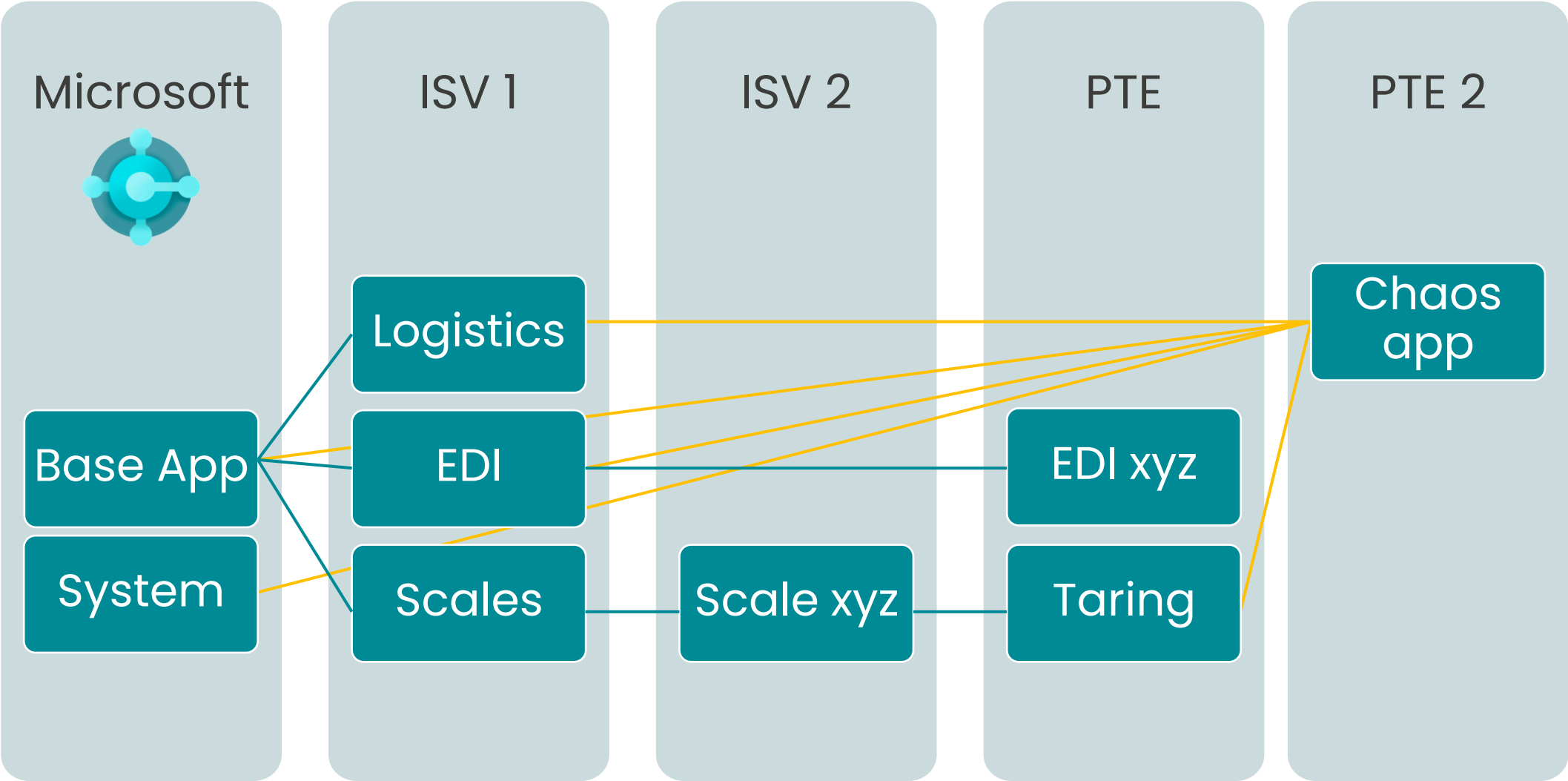
Upgrade-friendly:

- Architecture ensures continuous updates with minimal disruption
- Extensions keep customizations separate from core code



Extensibility

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Development Methodology

Business Central has its own language (AL), so it comes with all the benefits:...

- Source control
- Branching strategies
- CI/CD
- Automated tests
- ..

VAR & ISV



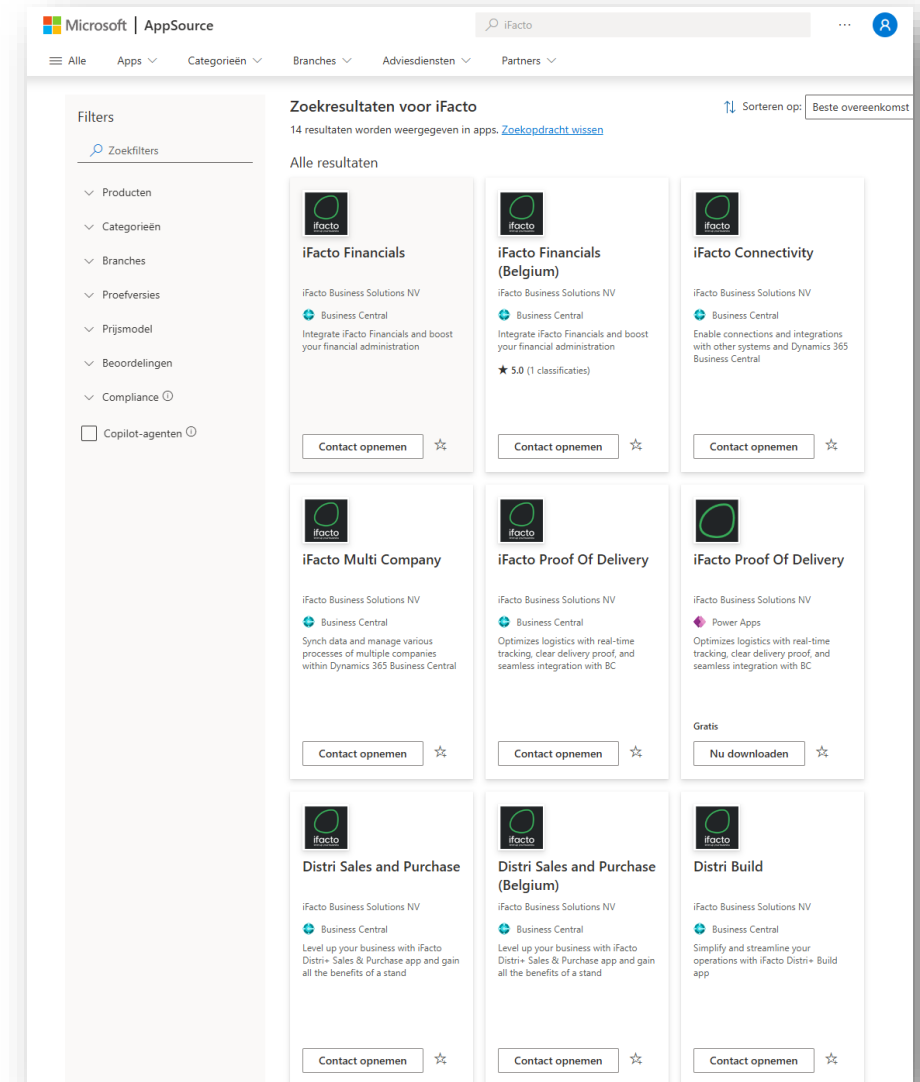
Very applicable in the BC market

ISV:

- Develop & Sell AppSource Apps
- Comply with new versions of BC

VAR:

- Implement BC and ISV Apps
- PTE Development



File Edit Selection View Go Run Terminal Help

EXPLORER

OPEN EDITORS

- CREASetupSubs.Codeunit.al BASE - App • Src\CR...
- MFFDataCheckNotificatSubs.Codeunit.al BASE - ...
- MFFPurchaseHeaderDataCheck.Codeunit.al BAS...
- app.json BASE - App
- DARDistriAdminRoleCenter.Page.al BASE - App • ...
- DARDistriAdmin.Profile.al BASE - App • Src\ DAR (...)

DISTRIPPS (WORKSPACE)

BASE - App

Src

- ICON
- II (Item Importer)
- ISC (ItemSearch Framework)
- JQM (Job Queue Manager)
- LEM (Last Error Mail)
- LOG (Logistics)
- LT (Linked Texts)
- MFF (Mandatory Fields)
- NTT (Number To Text)
- OP (Open Page)
- PRM (Parameter Tool)
- RDТ (Report Document text framework)
- REL (Release Mgt Framework)
- RH (Report Helper)
- SD (SystemData)
- TPL (Templates)
- TST (Test Framework)
- VMF (Validation Message Framework)
- BSEADMIN.PermissionSet.al
- BSEALL.PermissionSet.al
- BSEUSER.PermissionSet.al
- D365FULLACCESSExtBSE.PermissionSetExt.al
- FeatureLYCIFCExtBSE.EnumExt.al
- readme.md
- Translations
- .objidconfig
- app.json

OUTLINE

TIMELINE

AL OUTLINE

CREASetupSubs.Codeunit.al

MFFDataCheckNotificatSubs.Codeunit.al

MFFPurchaseHeaderDataCheck.C

BASE - App > app.json

```
1 {
2   "id": "70237ab8-4316-4404-bc9a-7f4c43bd5dbd",
3   "name": "iFacto Base",
4   "publisher": "iFacto Business Solutions NV",
5   "brief": "iFacto Base",
6   "description": "iFacto BASE app",
7   "version": "25.1.0.0",
8   "privacyStatement": "http://www.ifacto.be",
9   "EULA": "http://www.ifacto.be",
10  "help": "http://www.ifacto.be",
11  "contextSensitiveHelpUrl": "http://www.ifacto.be",
12  "url": "http://www.ifacto.be",
13  "logo": "Images\\AppLogo.png",
14  "platform": "25.0.0.0",
15  "application": "25.1.0.0",
16  "dependencies": [
17    {
18      "id": "2d1d1d89-8722-45af-a8f7-f2f8aa043767",
19      "publisher": "iFacto Business Solutions NV",
20      "name": "iFacto Environments",
21      "version": "25.1.0.0"
22    },
23    {
24      "publisher": "iFacto Business Solutions NV",
25      "name": "Distri Extended Tracking",
26      "id": "dc890dde-aed0-44e4-90ed-7b6c728e5b94",
27      "version": "25.1.0.0"
28    },
29    {
30      "id": "63c792ec-b3d2-47e5-b884-b7471dff2f4b",
31      "name": "iFacto Licensing",
32      "publisher": "iFacto Business Solutions NV",
33      "version": "25.1.0.0"
34    },
35    {
36      "id": "6ffdb5bc-fc7c-4137-9ee0-4bacc43f2da8",
37      "name": "iFacto Data Check",
38      "publisher": "iFacto Business Solutions NV",
39      "version": "25.1.0.0"
40    }
41  ],
42  "internalsVisibleTo": [
43    {
44      "id": "0f9d631a-8096-4a91-8a07-84b19e4a0225",
45      "name": "DISTRI-BASE-TEST"
```

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Power Platform

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Disclaimer



We could be completely wrong – but this is the world of PP from the eyes of a BC developer..

What is the Power Platform?



*A **low-code platform** to build **apps**, automate **workflows**, analyze **data**, and **integrate systems***

What is the Power Platform?

Microsoft Power Platform

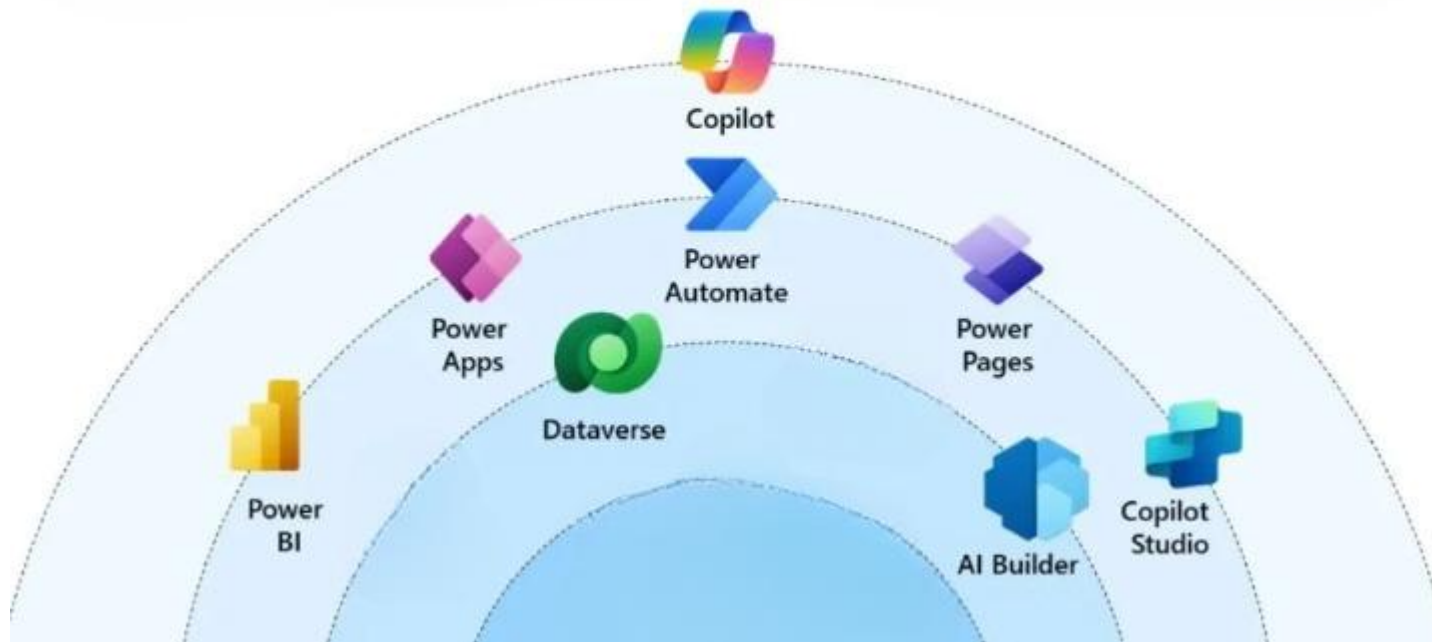
The world's **most complete** platform for developer productivity, spanning no-code to pro code



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Microsoft Power Platform

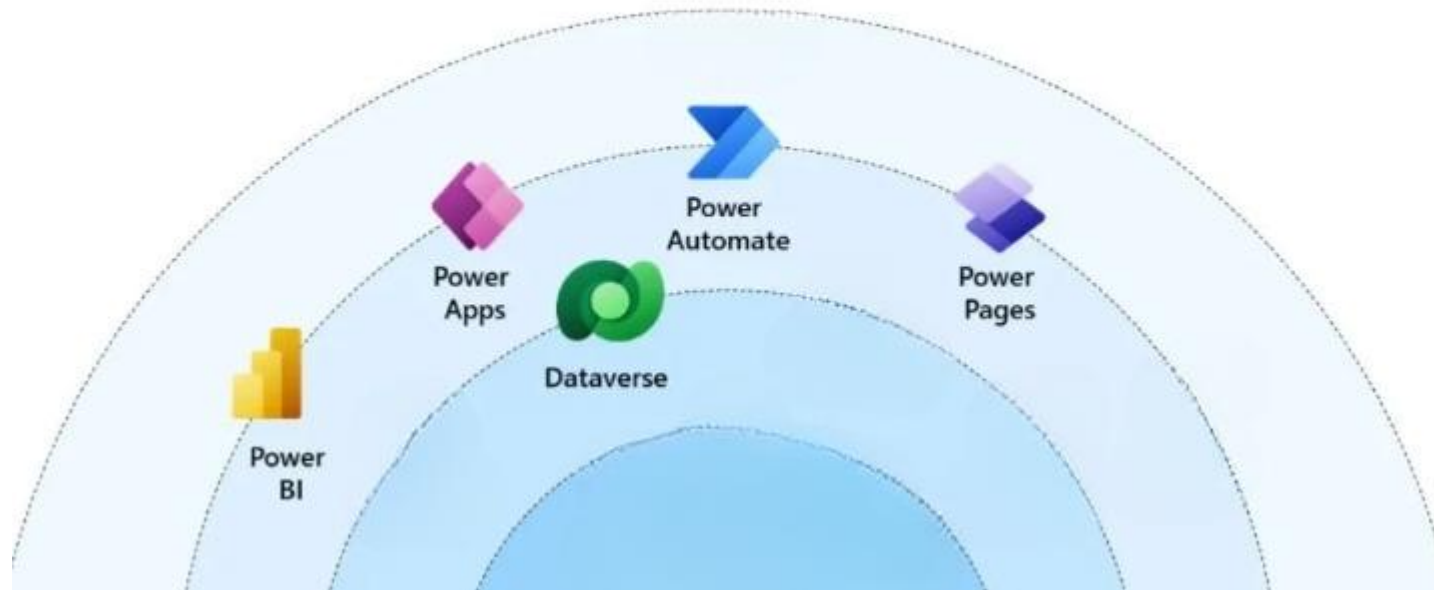
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Microsoft Power Platform

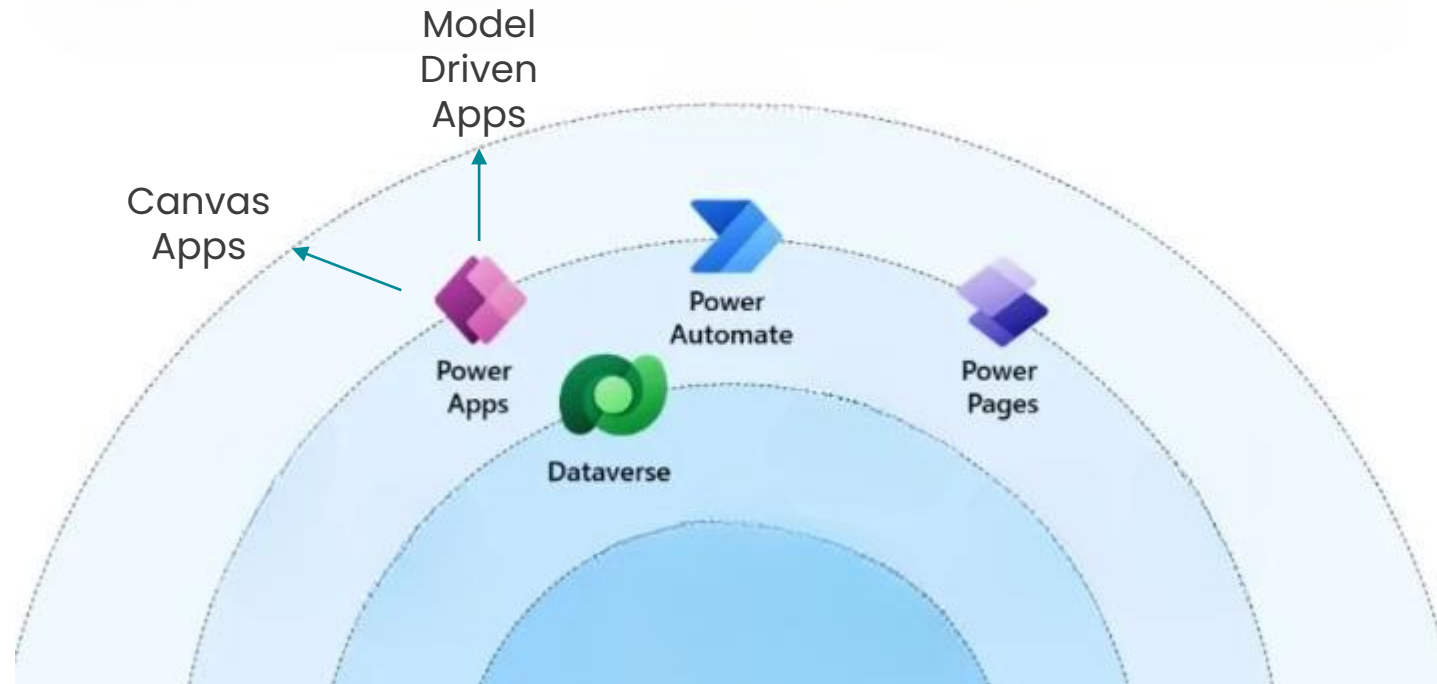
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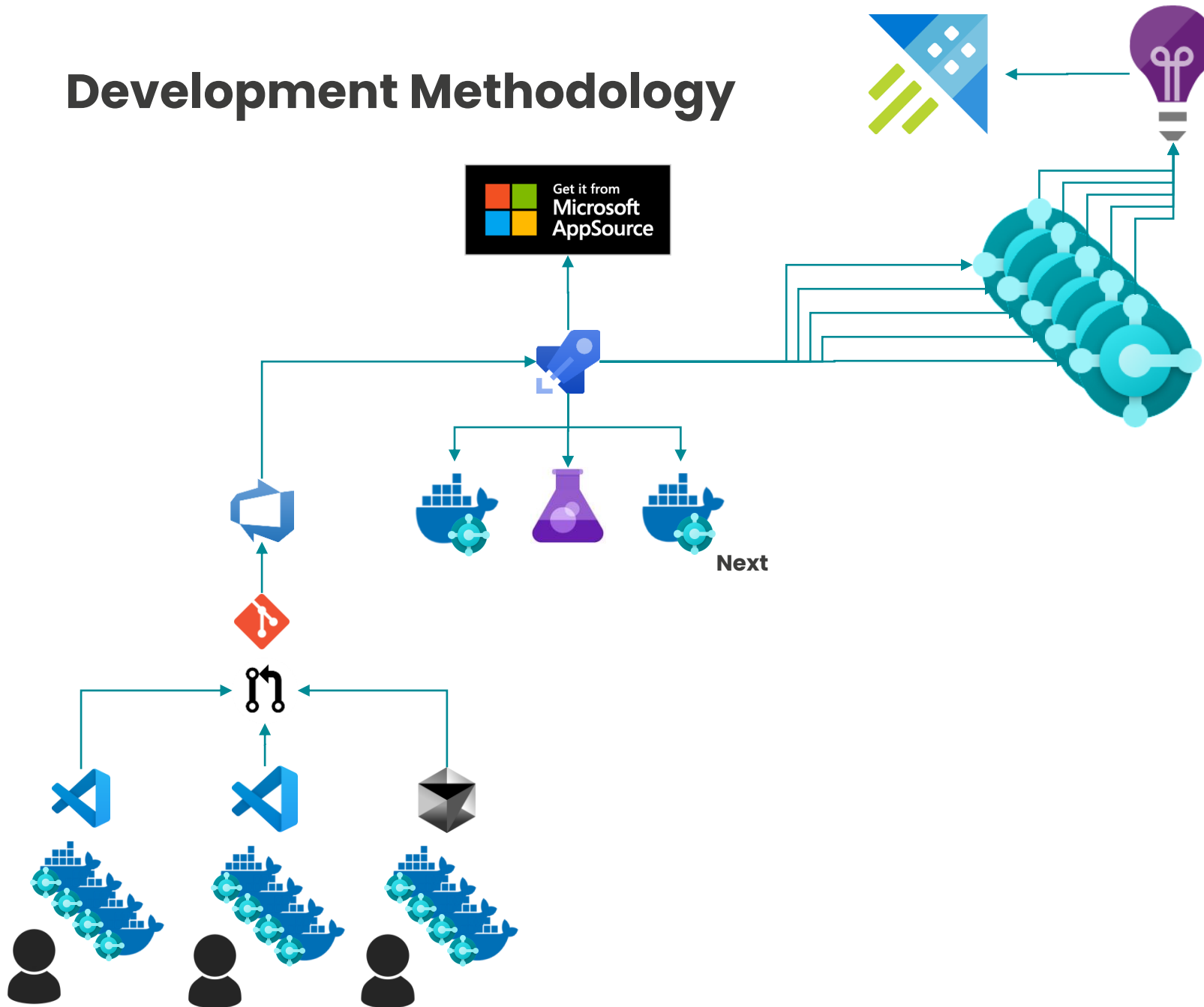
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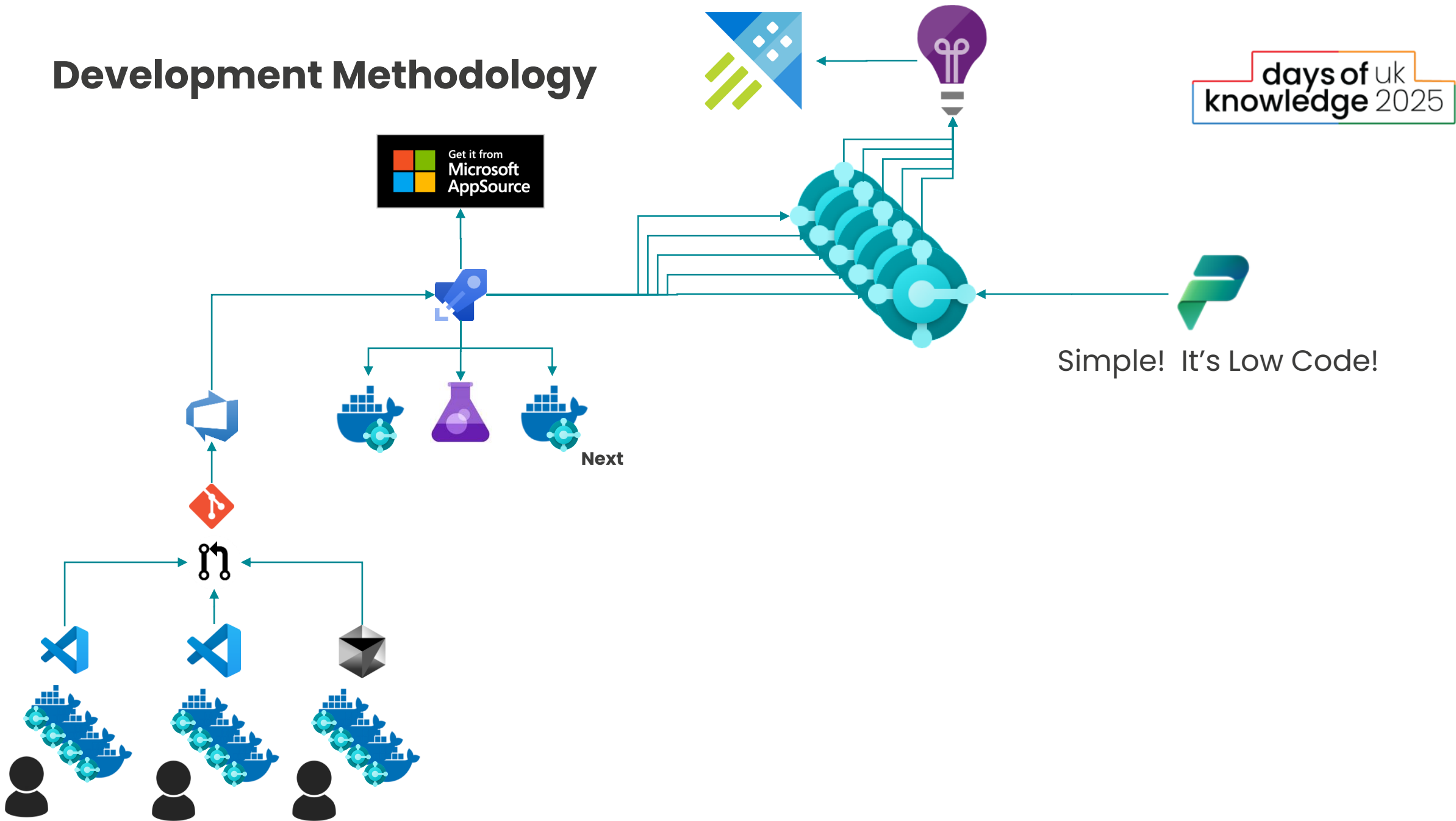
The world's **most complete** platform for developer productivity, spanning no-code to pro code



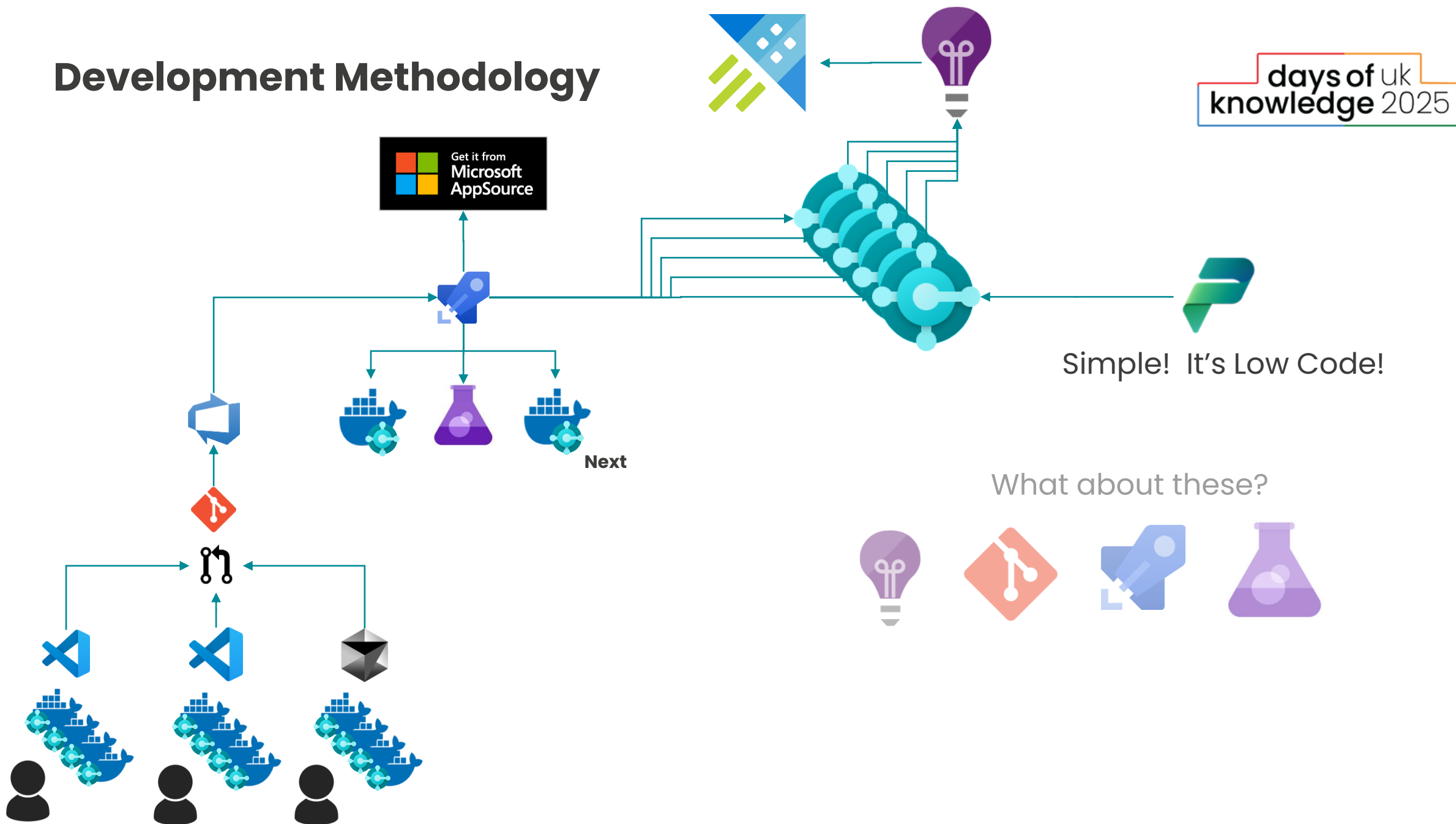
Development Methodology



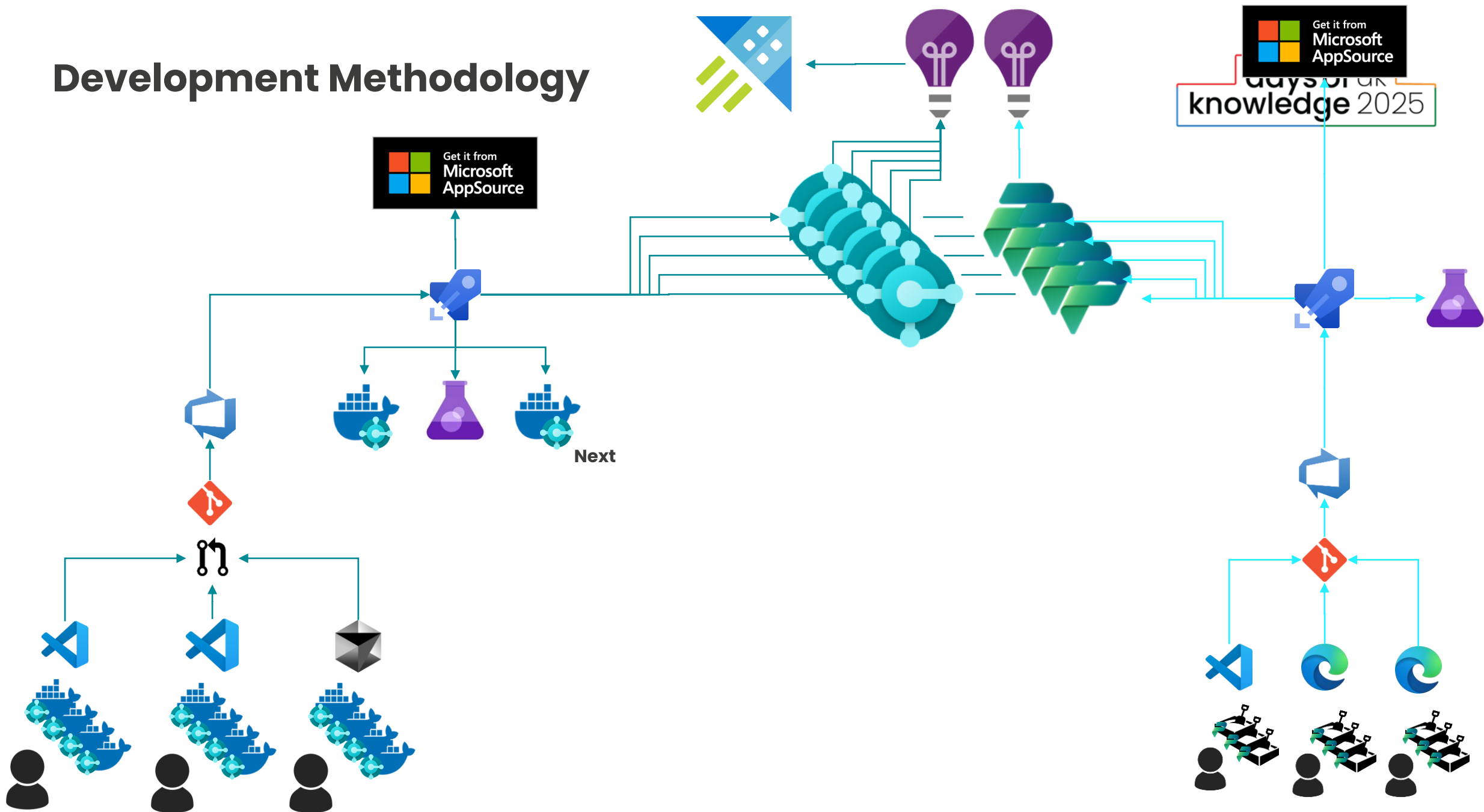
Development Methodology



Development Methodology

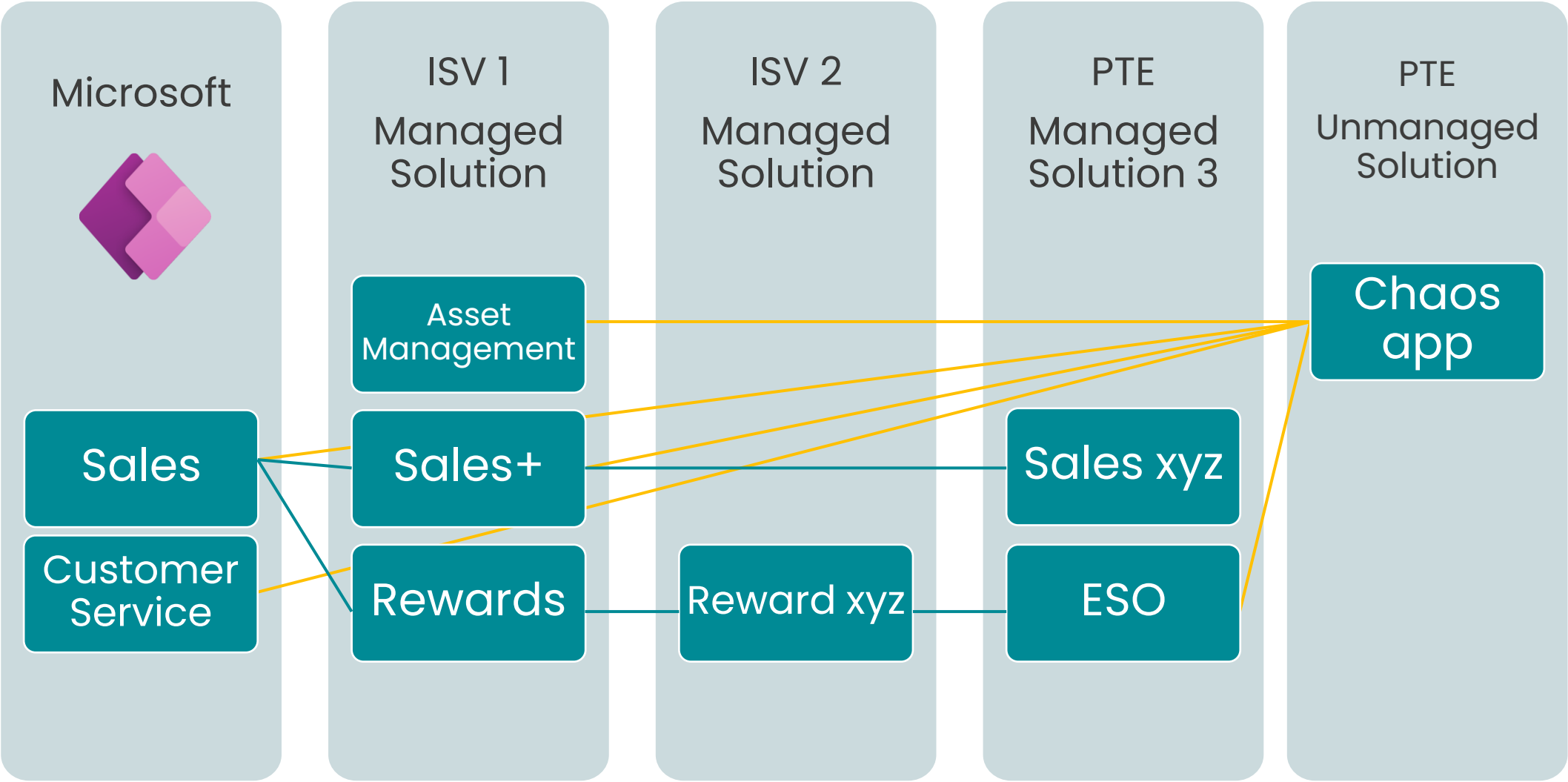


Development Methodology



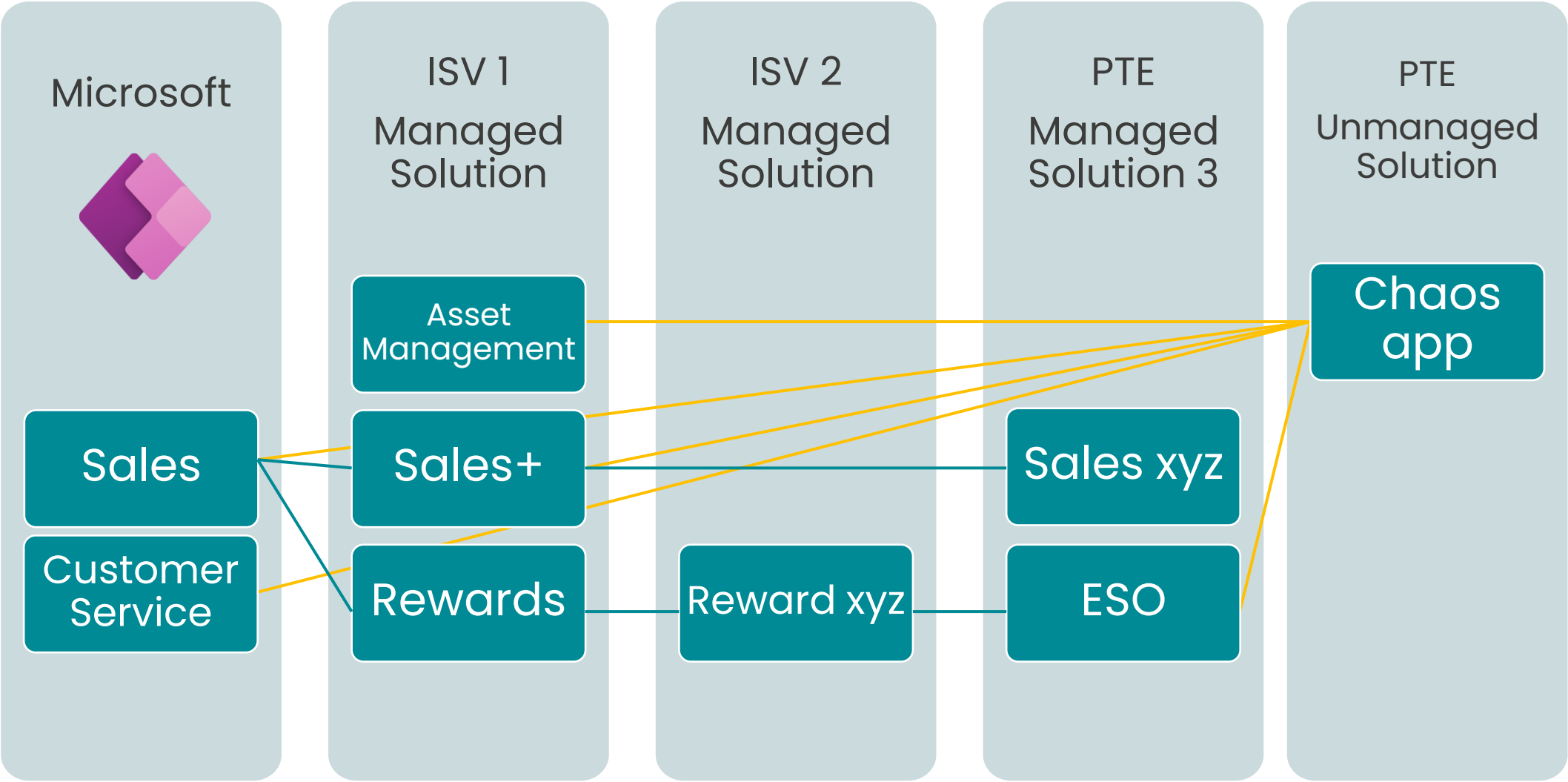
Extensibility

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Extensibility

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Extensibility



Chaos
app

Extensibility



Canvas
app

Citizen Developer

What?

Non-professional dev

Typically a business user

Typically outside traditional dev teams

Low Code/No Code solutions



Citizen Developer

"I've built a very complex set of Power Automate flows that became the backbone of our Sales department.

When I left, it all fell apart"

-Tine Starič

Citizen Developer

SMB Space

Resources

Minimal IT support
Relies on intuitive tools

Best Practices

Rarely applied
focus on quick fixes
inconsistent solutions
inefficient solutions.

Conclusion

just doesn't work

Enterprise

IT department
governance policies
professional developers

Consistently applied
Care about maintainability, security, and
alignment (with Business Central).

Conclusion

Could work

When you combine PP with BC...

Deployment

You can't combine the installation of both apps

- Either BC first, but what if Power Platform fails?
- Or PP first, but what if BC fails?

Rollback

BC: not possible to install previous version.

PP: IS possible to roll back

Extensibility

Business Central:

Extensibility platform

Power Platform:

PTE apps in Power Platform don't exist.

Layer Merging does exist for Model Drive Apps.

Canvas Apps: no extensibility.

Conclusion: PP has an extensibility story, but not from how we use and see it.

Validation



No automated validation/tests possible.

Now just imagine the monthly updates..

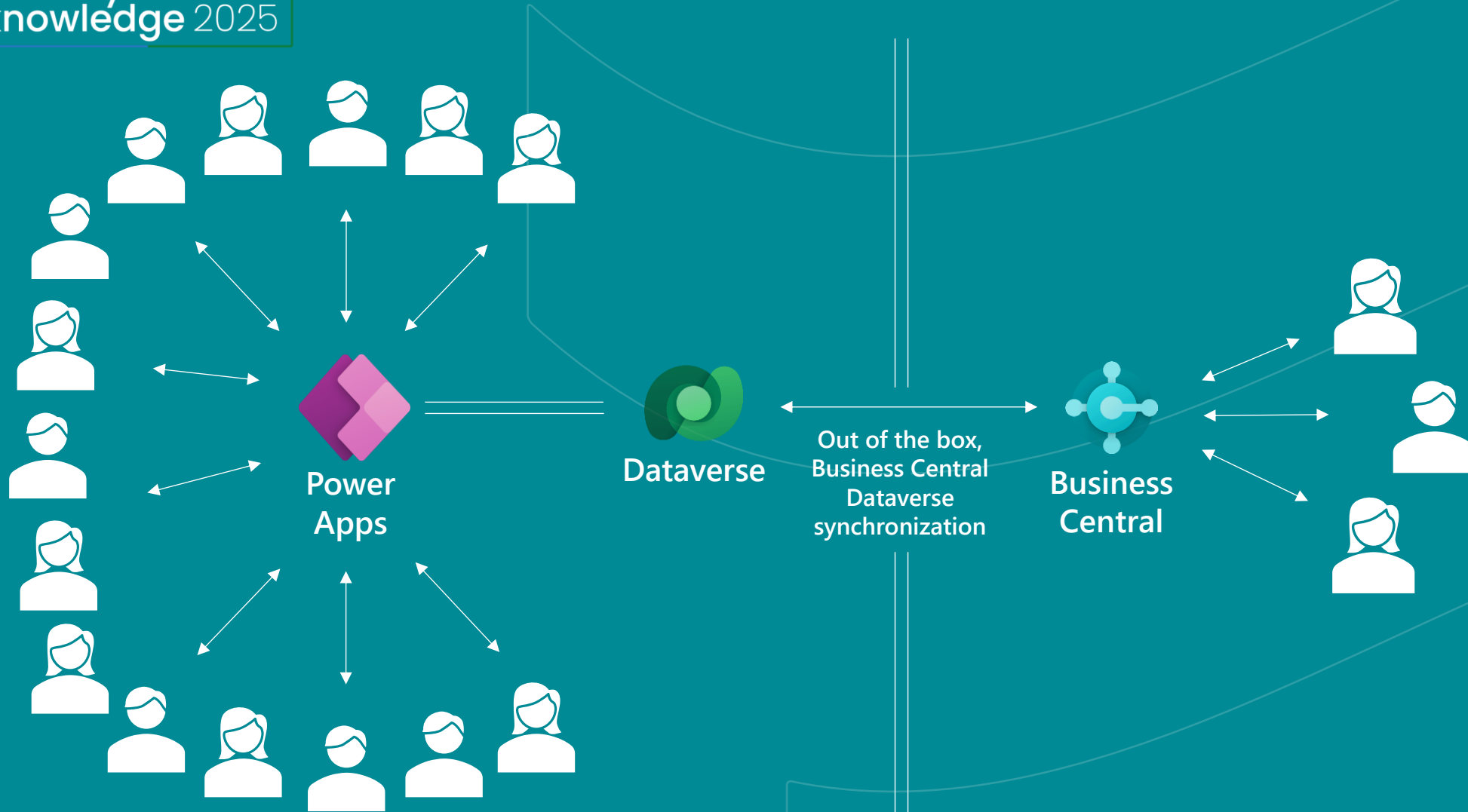
Licensing differences

Multiplexing

The use of hardware or software to pool connections, reroute or indirectly access information, and/or reduce the number of devices or users that directly (or indirectly) access or use a product.

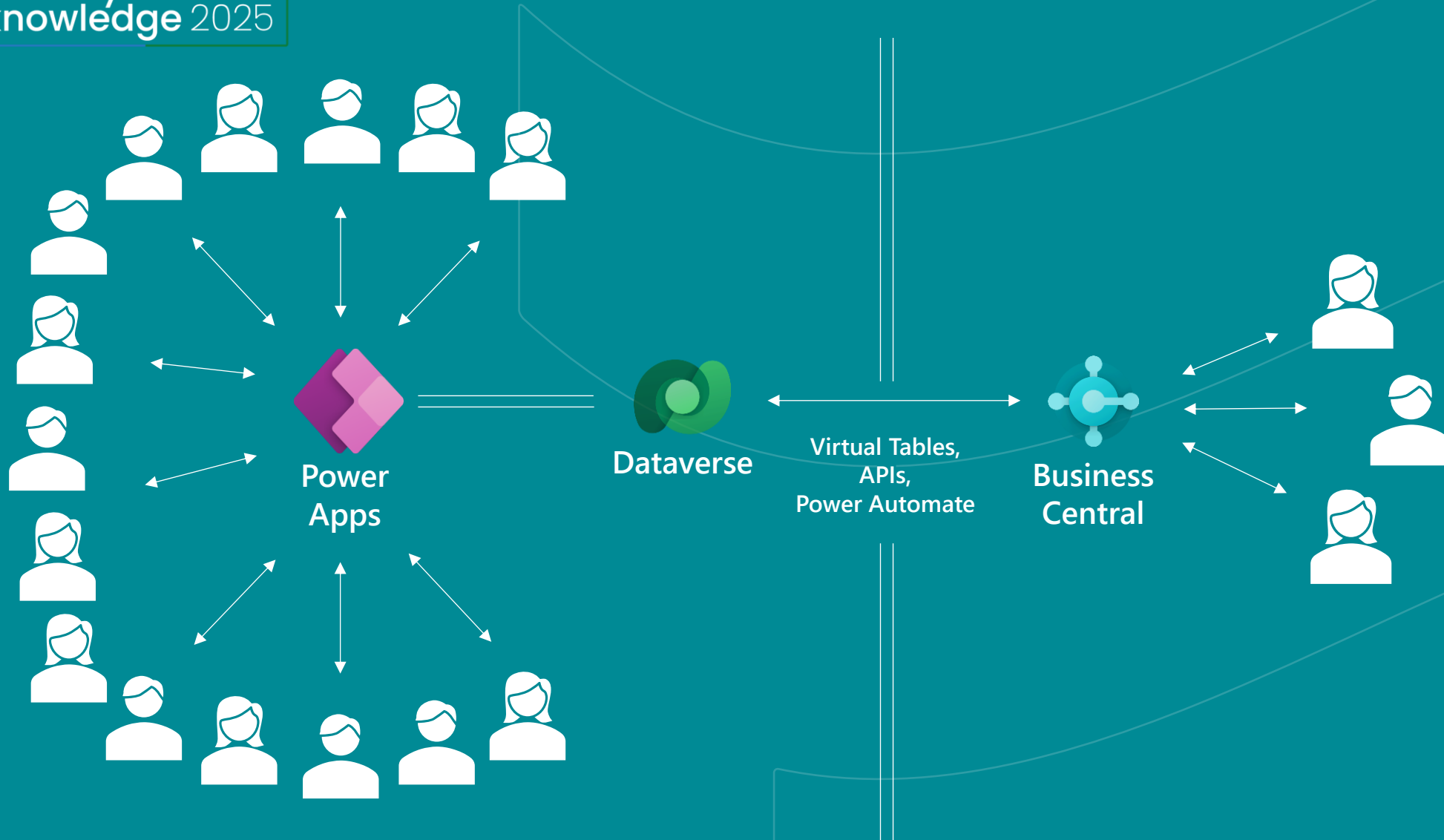
Multiplexing

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Multiplexing

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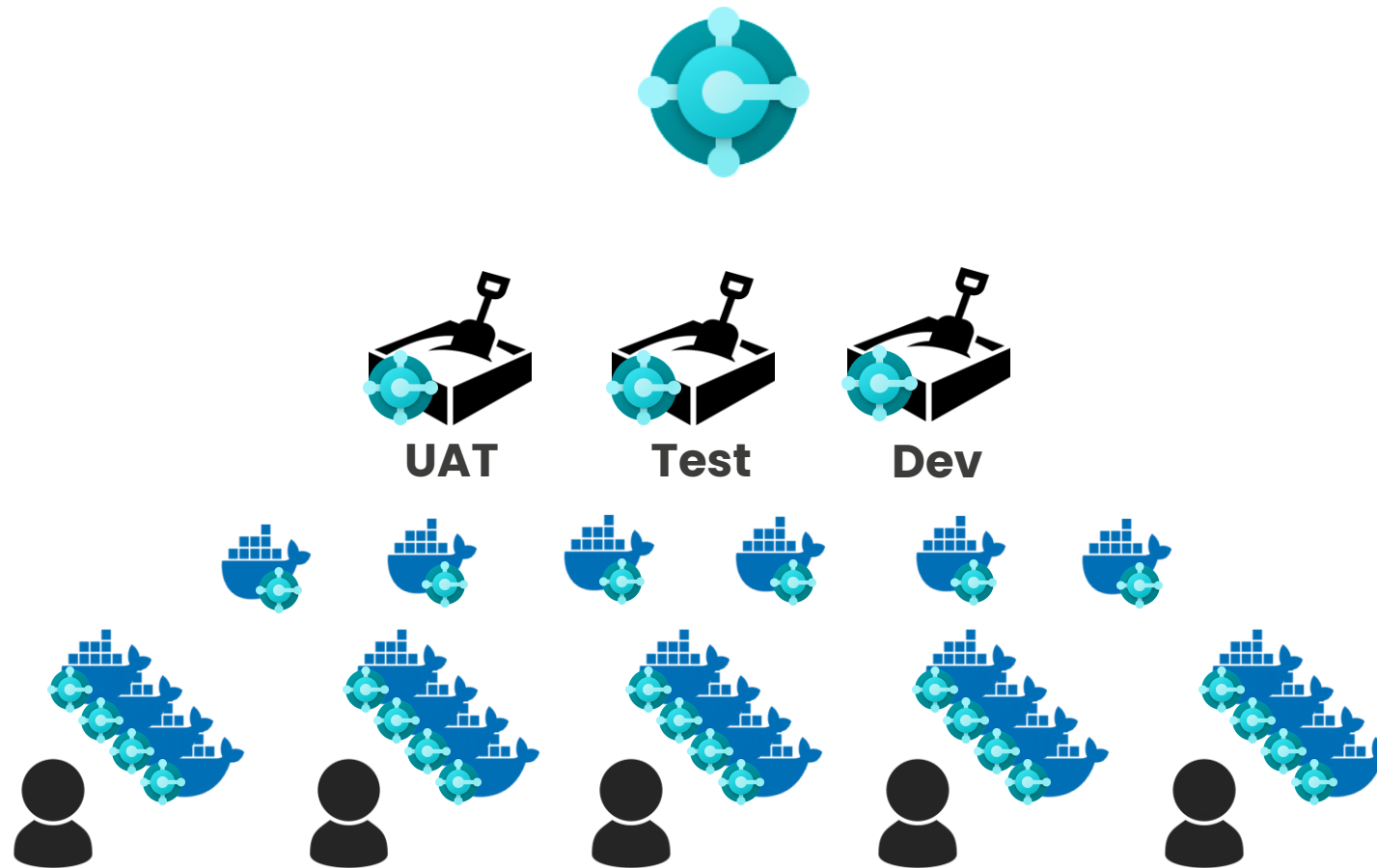


Development Environments

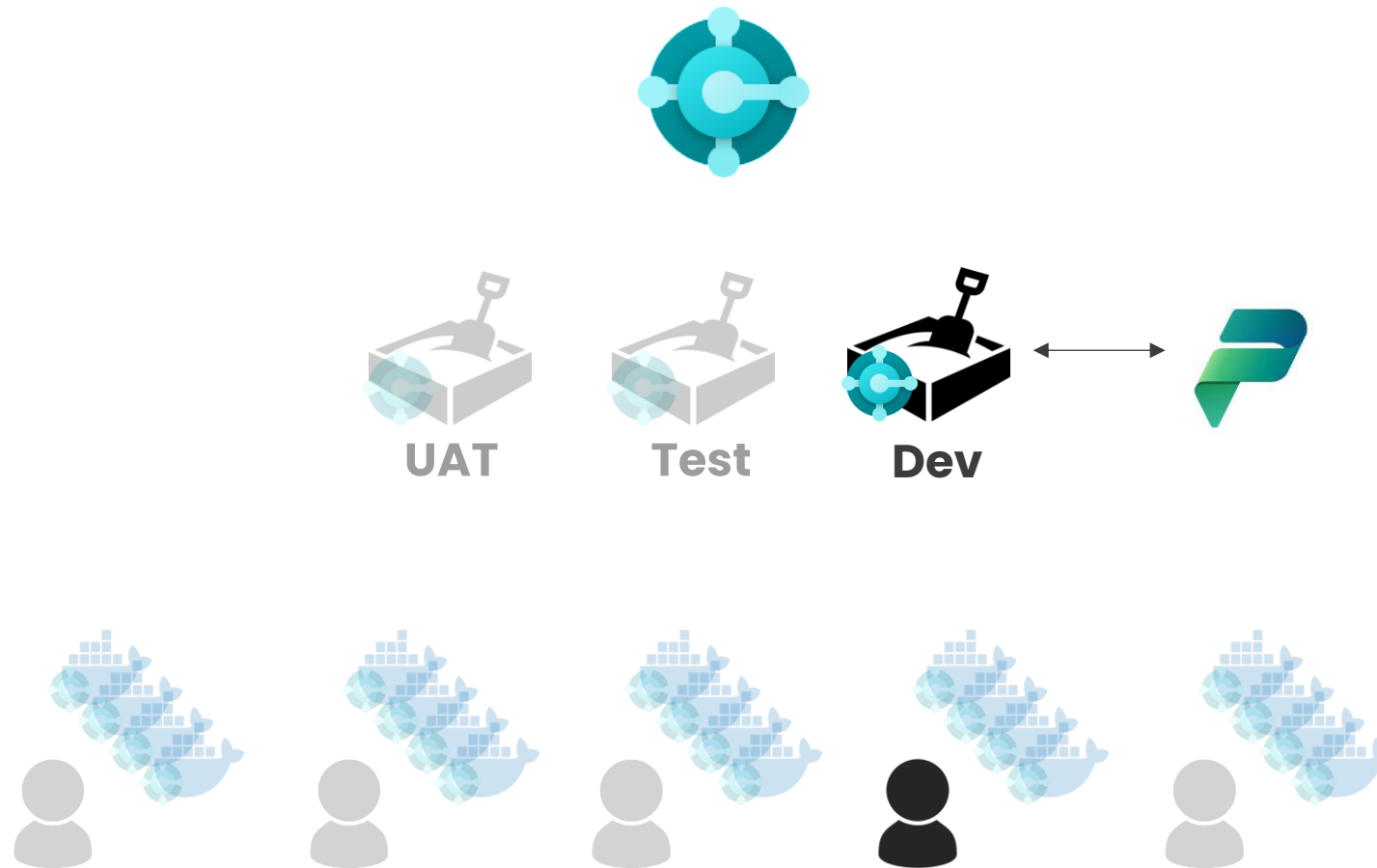
Things to take into account:

- BC only has 3 sandboxes + 1 production
- Power Platform: as many environments as you want
- BC has multiple companies in one environment, PP “expects” one

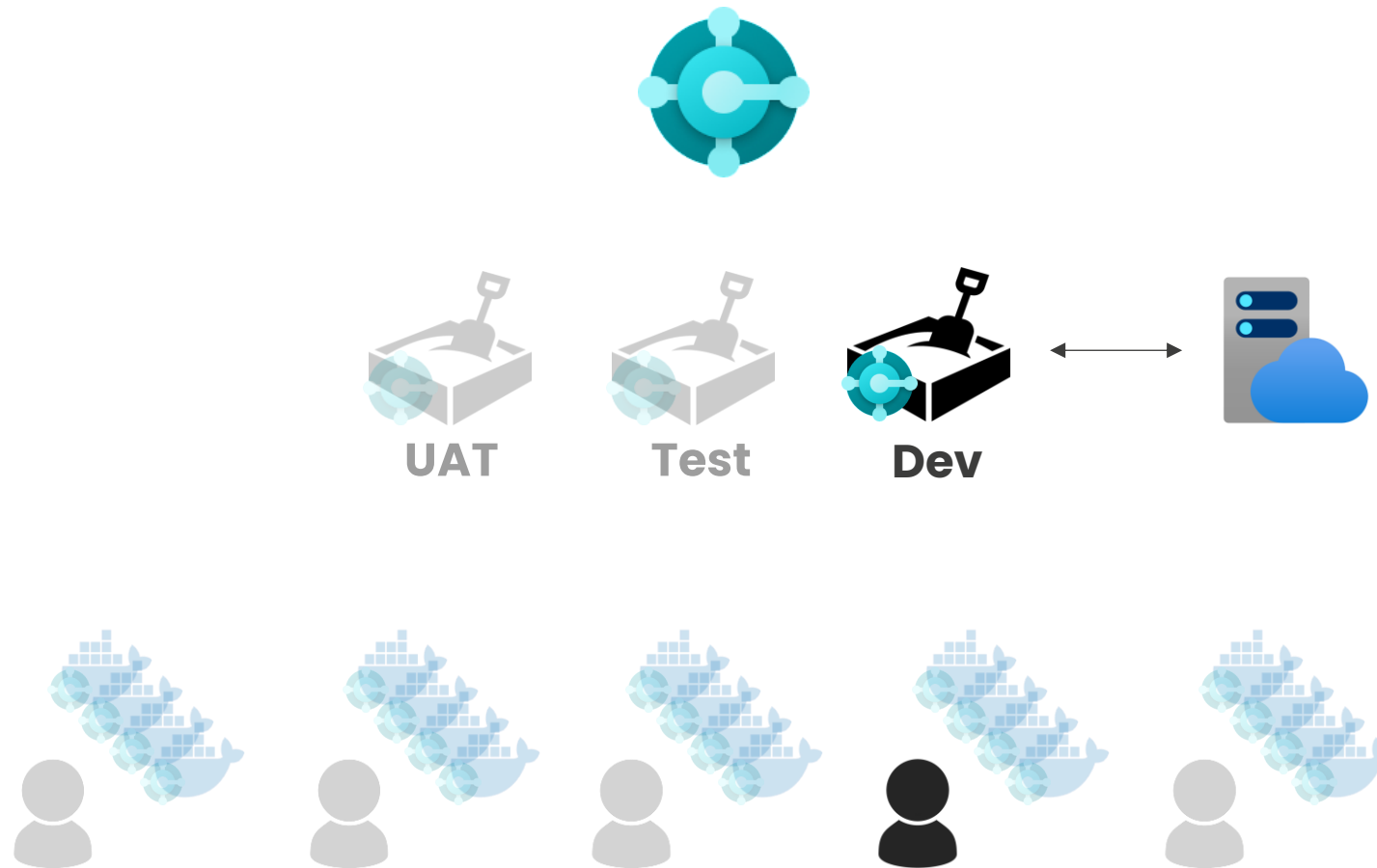
Environments – BC



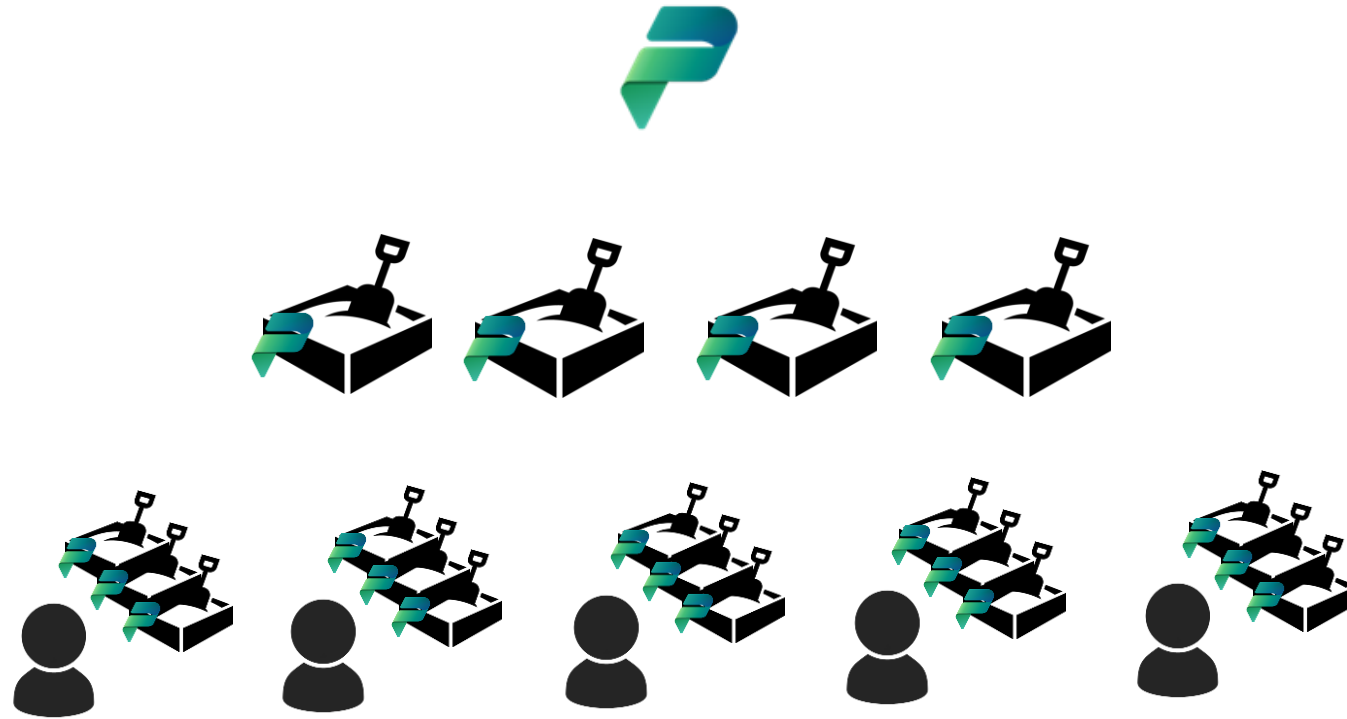
Environments – BC



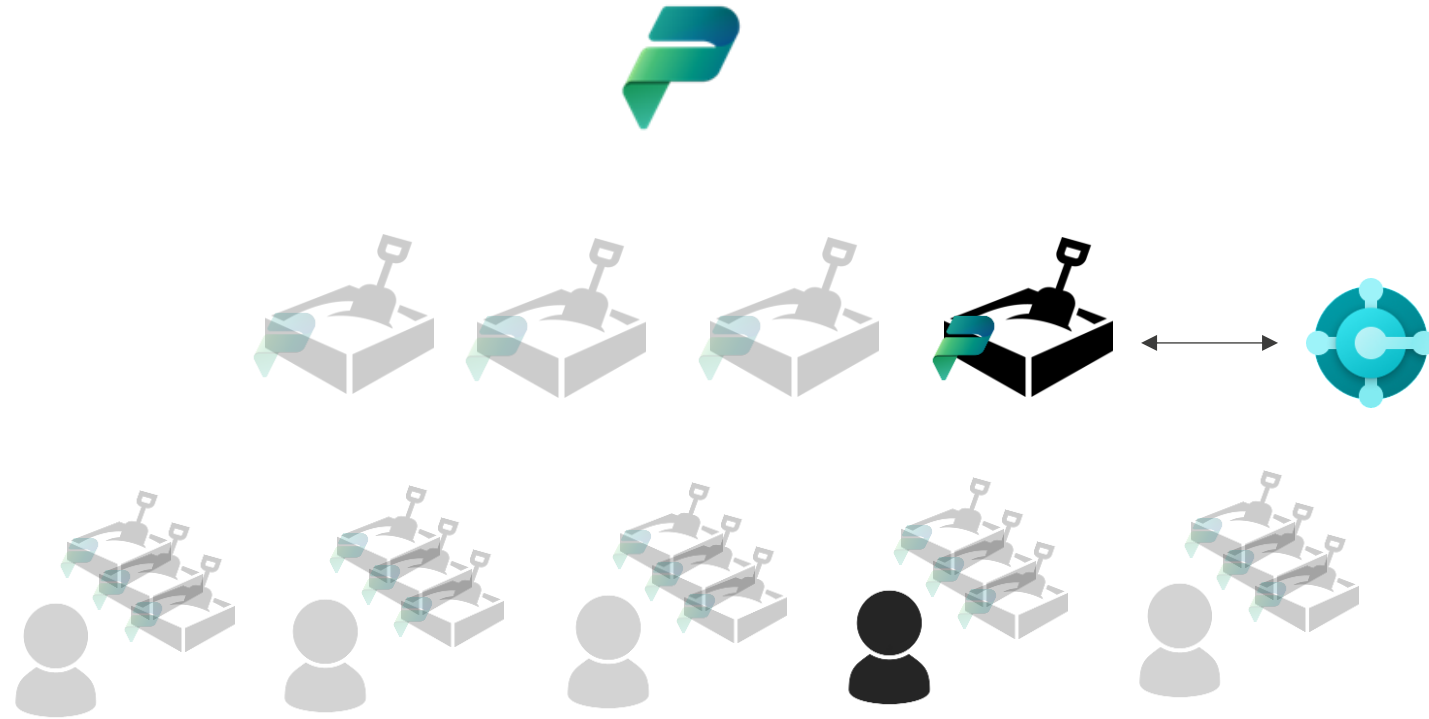
Environments – BC



Environments – PP



Environments – PP



Data(verse)



BC stores critical business data (customers, sales, inventory, ...)

To work with the data, takes effort:

- Use Power Platform Connectors to work with BC APIs
- The BC is not “just” in Dataverse. You can:
 - Use BC Sync
 - Virtual Tables

Dataverse & BC

BC Sync:

BC copies data from BC to Dataverse and back

- Merge Conflicts
- Bidirectional
- Not real time
- Outside-the-box = development

Virtual Tables:

- BC API = development

**Just because you can,
doesn't mean you should**

**Just because you can 't,
means you should**

Power Automate

Triggers are awesome

AL cannot “hook” to webhooks natively



Power
Automate
Logo



Forms Logo

Power Automate

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```
5 references
table 50100 "Teams Setup"
{
    Caption = 'Teams Setup' page 50100 "Teams Setup"
    DataClassification = {
        PageType = Card;
        ApplicationArea = All;
        UsageCategory = Administrative;
        SourceTable = "Teams Setup";
        InsertAllowed = false;
        DeleteAllowed = false;
        Caption = 'Teams Setup';
    }
    fields
    {
        2 references
        field(1; "Primary")
        {
            Caption = 'Primary'
            DataClassification = {
                PageType = Card;
                ApplicationArea = All;
                UsageCategory = Administrative;
                SourceTable = "Teams Setup";
                InsertAllowed = false;
                DeleteAllowed = false;
                Caption = 'Teams Setup';
            }
        }
        5 references
        field(2; "Client")
        {
            Caption = 'Client'
            DataClassification = {
                PageType = Card;
                ApplicationArea = All;
                UsageCategory = Administrative;
                SourceTable = "Teams Setup";
                InsertAllowed = false;
                DeleteAllowed = false;
                Caption = 'Teams Setup';
            }
        }
        0 references
        trigger OnValueChanged
        begin
            if xRec.NewRecord
            then
                SetField(Enabled, true);
            end;
        end;
    }
    layout
    {
        area(Content)
        {
            group(General)
            {
                Caption = 'General'
                field(Enabled; "Enabled")
                {
                    ApplicationArea = All;
                    ToolTip = 'Specifies whether the team is enabled.';
                }
            }
            group(Authentication)
            {
                Caption = 'Authentication'
                field(Enabled; "Enabled")
                {
                    ApplicationArea = All;
                    ToolTip = 'Specifies whether the team is enabled.';
                }
            }
        }
    }
}

1 reference | 0% Coverage
local procedure GetAccessToken() Success: Boolean
var
    RequestUrl: Text;
    ResponseText: Text;
    FormContent: Text;
    FormContentLines: Text;
    JsonObject: Text;
    ClientId: Text;
    ClientSecret: Text;
    TenantId: Text;
begin
    if not TeamEnabled
    then
        Error('Team is not enabled.')
    end;
    // Get credentials
    ClientId := GetTeamClientId();
    ClientSecret := GetTeamClientSecret();
    TenantId := GetTeamTenantId();

    if (ClientId = '' or ClientSecret = '' or TenantId = '')
    then
        Error('Missing team credentials.')
    end;

    RequestUrl := GetTeamAuthUrl();
    ResponseText := GetAccessToken(RequestUrl, ClientId, ClientSecret, TenantId);
    if ResponseText = ''
    then
        Error('Failed to get access token.')
    end;
    Success := true;
end;

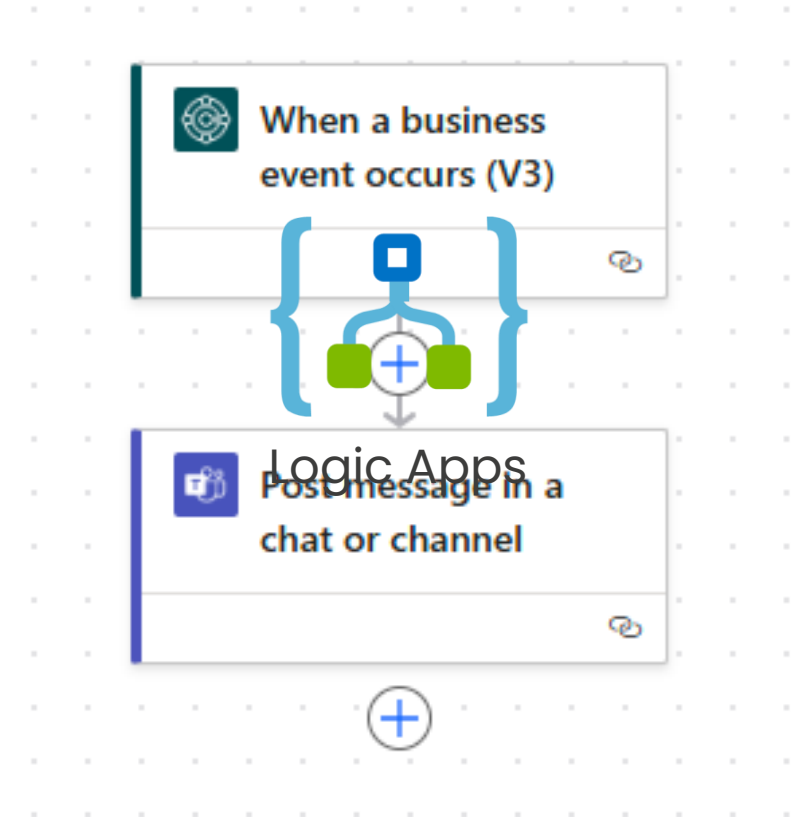
2 references | 0% Coverage
procedure SendChannelMessage(ChannelId: Text; MessageText: Text)
var
    RequestBody: Text;
    ResponseText: Text;
begin
    if not EnsureTeamEnabled()
    then
        Error('Team is not enabled.')
    end;
    // Create request
    HttpRequest.Method(MethodPost);
    HttpRequest.SetRequestUri(GetTeamAuthUrl());
    HttpRequest.GetHeaders(HttpHeaders);
    // Format the request body
    RequestBody := FormatMessage(MessageText);
    MessageText := SendRequest(RequestUrl, ClientId, ClientSecret, TenantId, RequestBody);
    Subject := GetTeamSubject();
    // Set headers
    HttpHeaders.Remove('Content-Type');
    HttpHeaders.Remove('Accept');
    HttpHeaders.Add('Content-Type', ContentTypeJson);
    HttpHeaders.Add('Accept', AcceptJson);
    HttpHeaders.Remove('Authorization');
    HttpHeaders.Add('Authorization', StrSubstNo('Bearer %1', ResponseText));
    // Set body if provided
    if MessageText <> ''
    then
        HttpRequest.SetRequestBody(MessageText);
    end;
    SendRequest(RequestUrl, ClientId, ClientSecret, TenantId, RequestBody);
end;
```

Power Automate

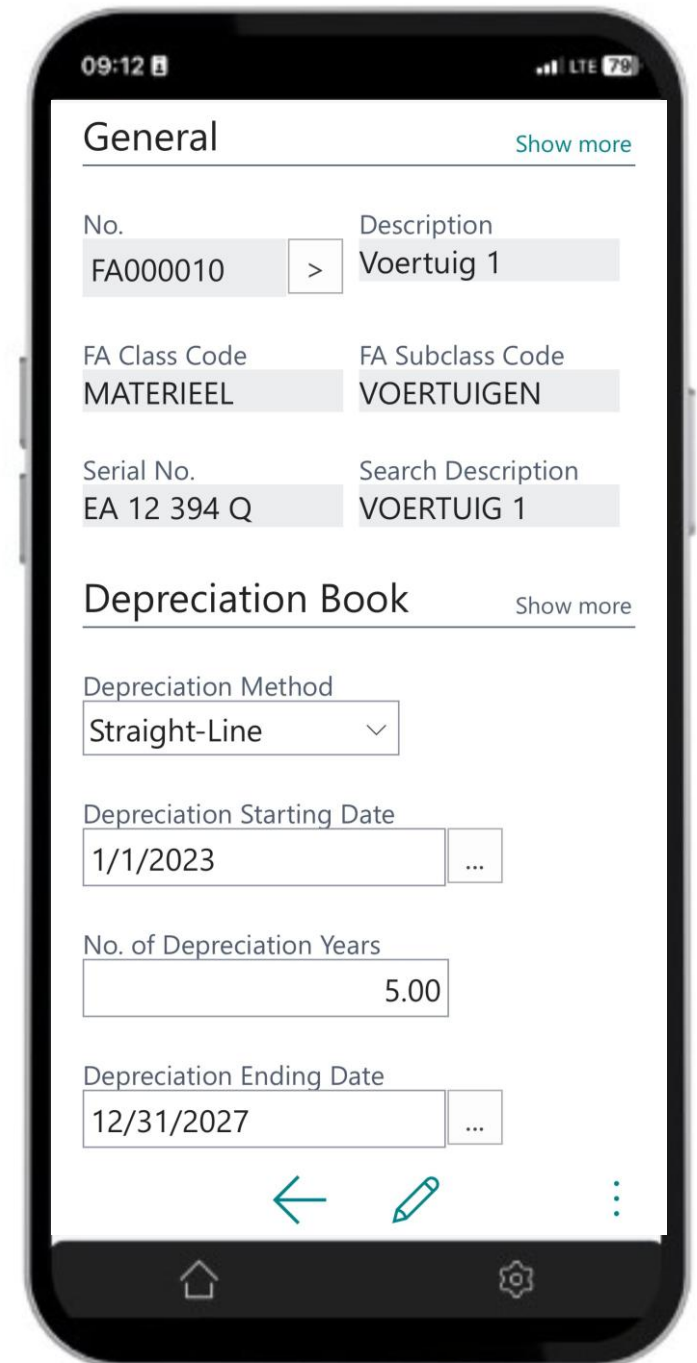
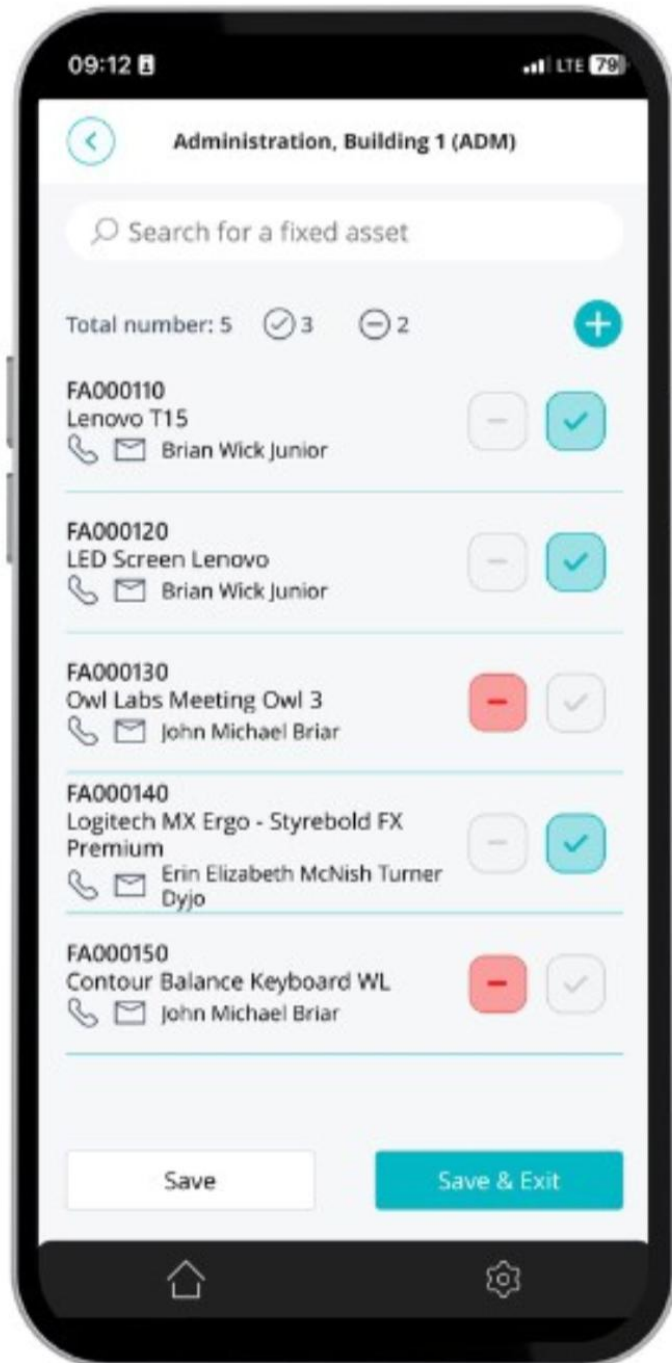
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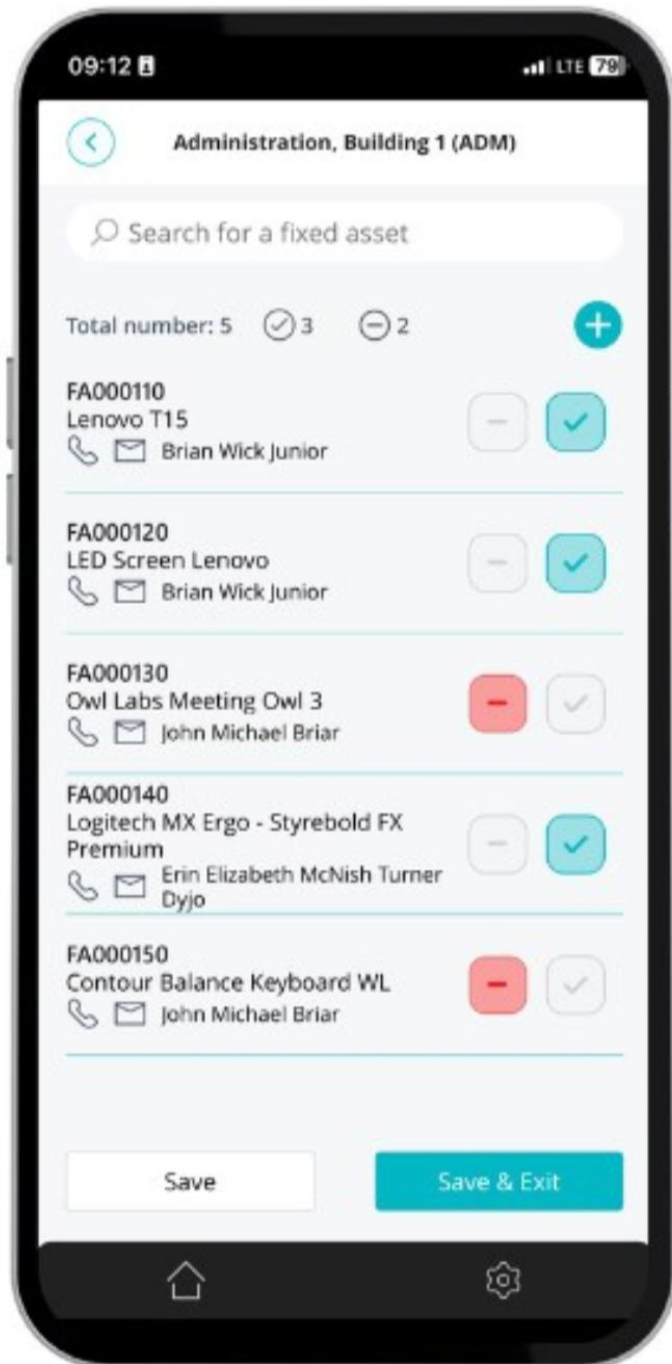
Simplify non-critical integrations

It's Power Automate not Power Integrate 🤖

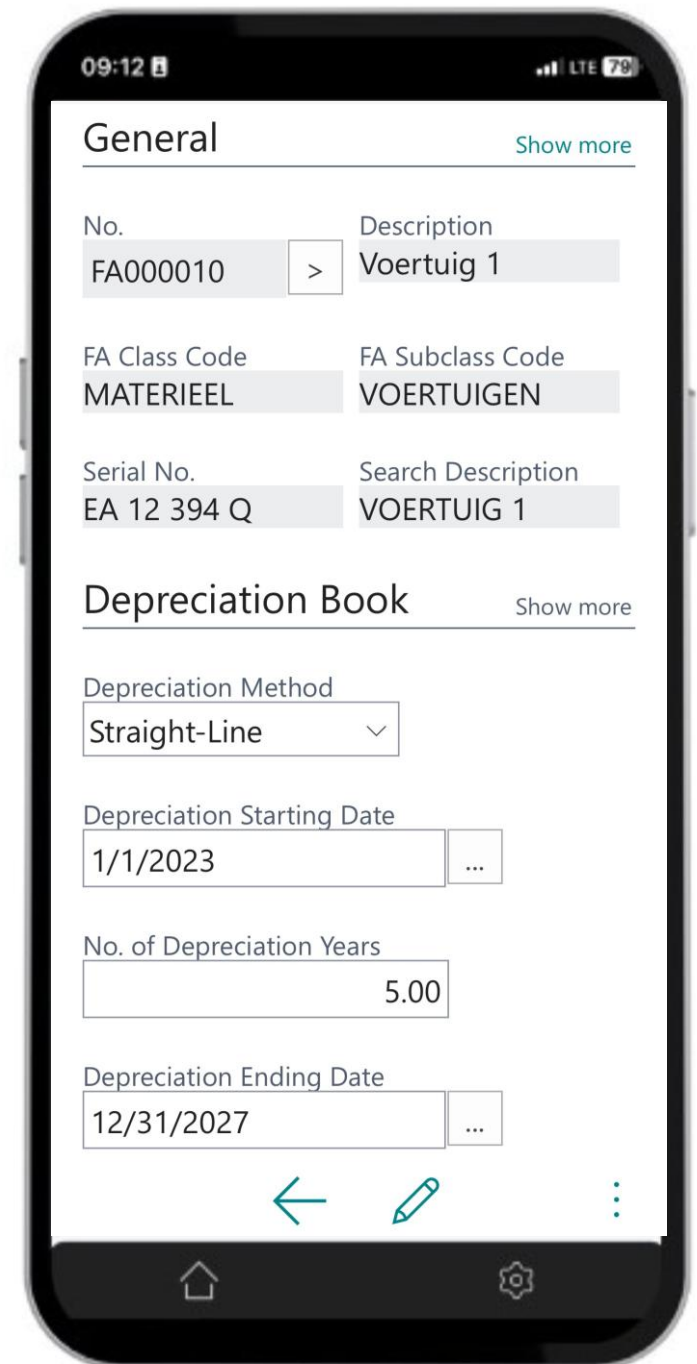


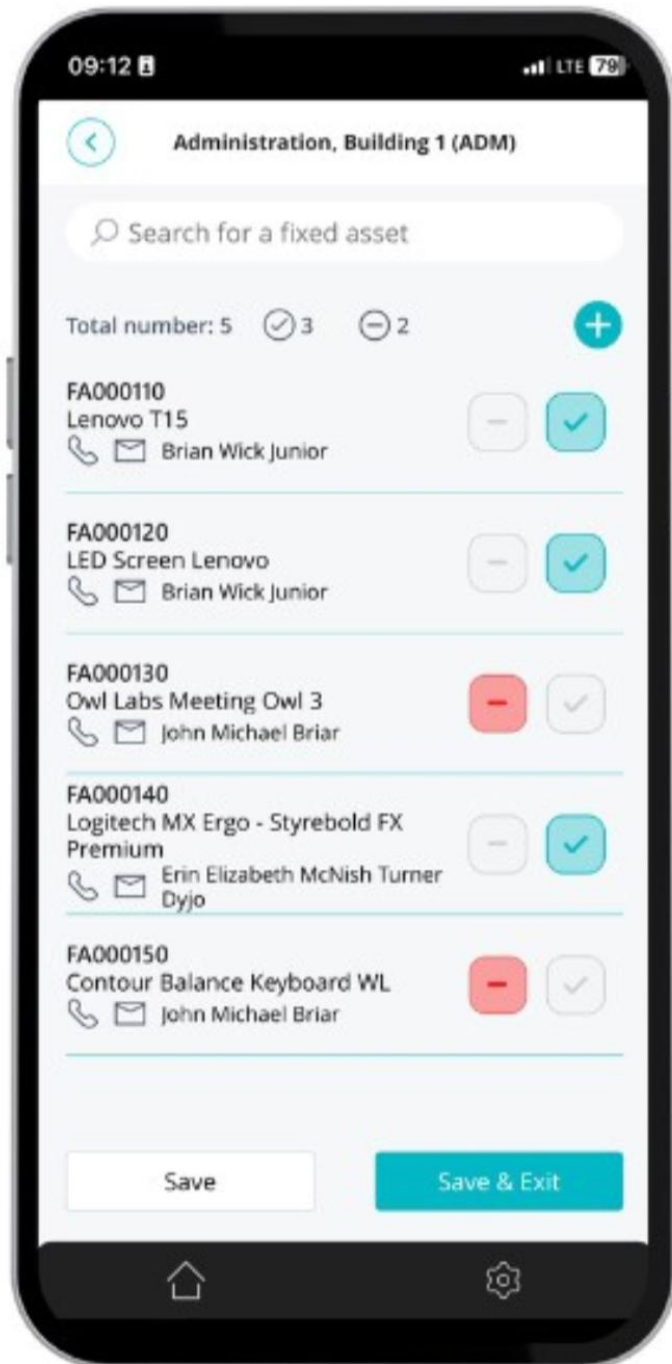
Simplistic UI



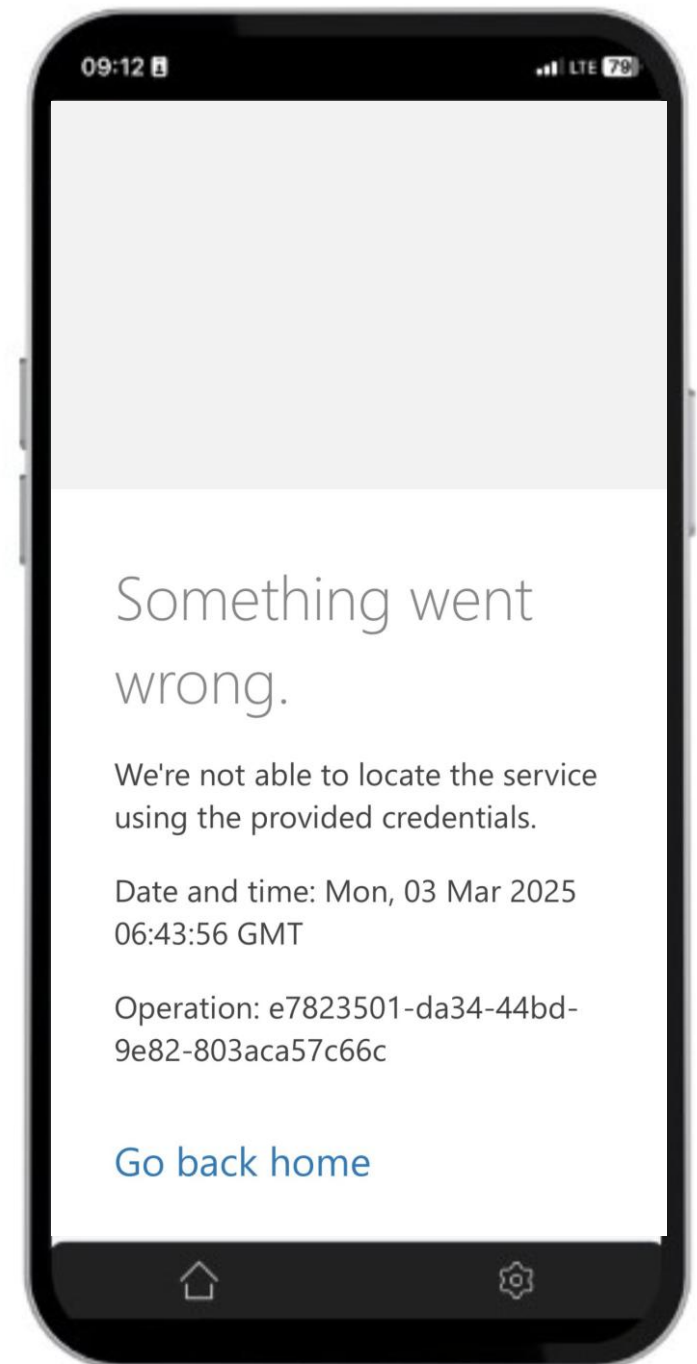


Offline Mode





Offline Mode



Power Pages



Home | Pages | Contact us | Search | Sign in

Item Catalog

See a list of all our wonderful items that will make your offices shine in a completely new light!

Product ID	Name	Description	Quantity On Hand	List Price
1896-S	ATHENS Desk		4.00	USD649.40
1900-S	PARIS Guest Chair, black		0.00	USD125.10
1906-S	ATHENS Mobile Pedestal		5.00	USD281.40
1908-S	LONDON Swivel Chair, blue		3.00	USD123.30
1920-S	ANTWERP Conference Table		10.00	USD420.40
1925-W	Conference Bundle 1-6		0.00	USD122.50
1928-S	AMSTERDAM Table		0.00	USD35.60

ATHENS Desk

The only option for truly sky-rocketing your productivity. The perfect addition to your home office.

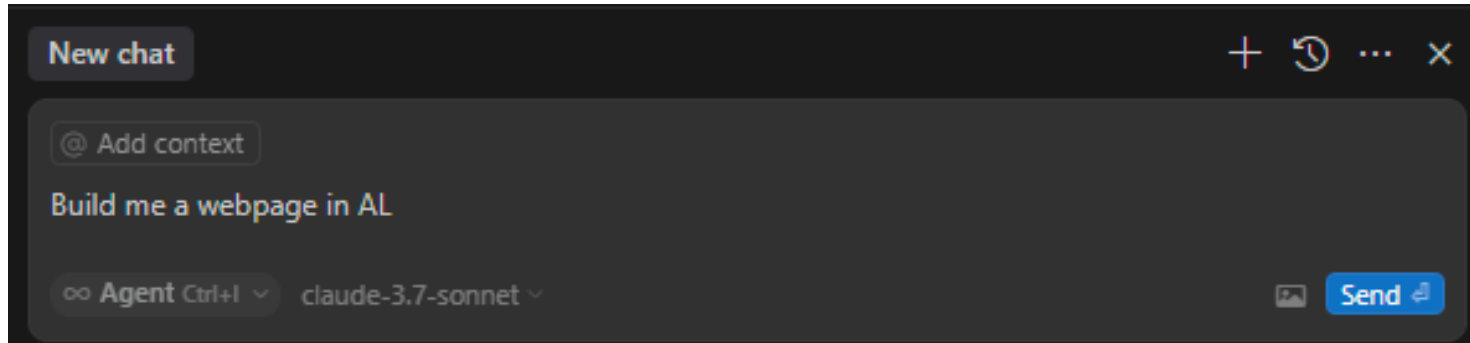
PARIS Guest Chair

Seating experience reimagined. You will never want to get up ever again.

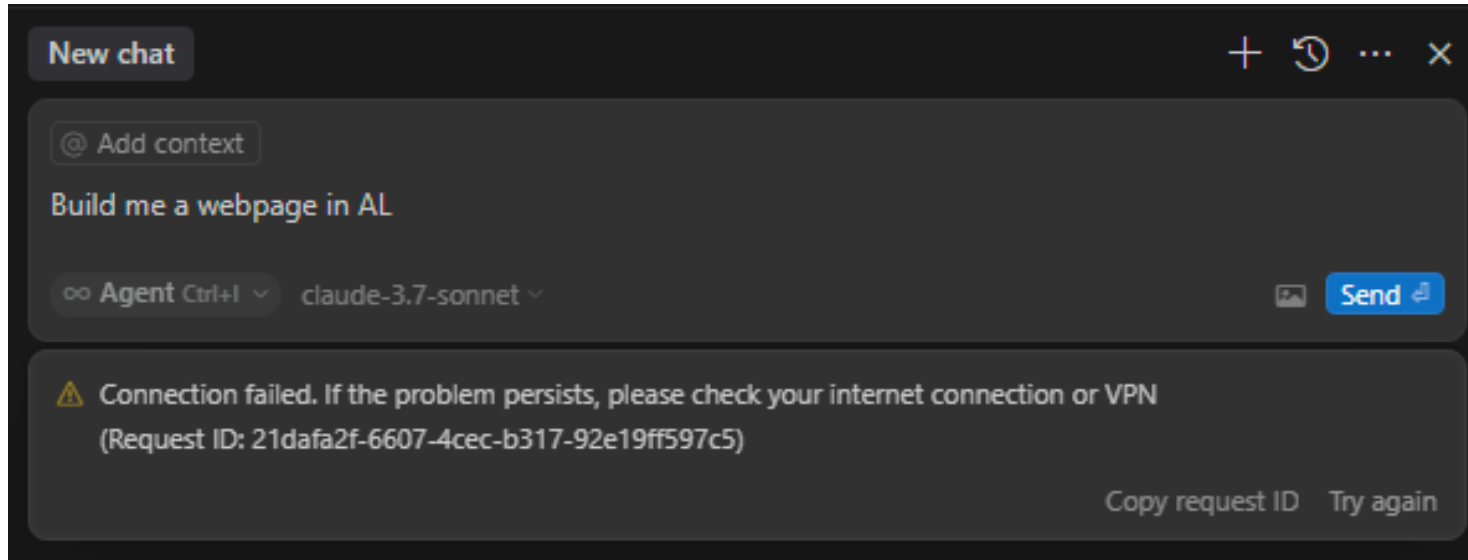
ATHENS Mobile Pedestal

The perfect addition to any balcony. Give the people an option to put that perfect cup of coffee down for a second and enjoy the morning air.

Power Pages



Power Pages



Model-Driven Apps



I honestly don't have a good example. It was always better to build it in BC

Unless you're supporting a process that has nothing to do with BC

You don't "*just do Power Platform*" without making a mess

You don't "*just do Business Central*" without making a mess

It makes sense when a use case is very difficult or impossible in BC

Low code $\neq$ Free

Low code $\neq$ Low maintenance

Low code $\neq$ Bad solution

**Just because you can,
doesn't mean you should**

**Just because you can 't,
means you should**

Give us Feedback!

and help us improve!

Please rate my session in the Conference App!

and leave a constructive comment.

Thank you !

